

Nicolas Chopin

Professor of data sciences

ENSAE
5 avenue Henry Le Chatelier
91764 Palaiseau Cedex FRANCE
☎ +33 1 41 17 65 22
✉ nicolas.chopin@ensae.fr
🌐 <https://nchopin.github.io/>



Experience

- Sept. 2006–
to date **Professor of data sciences**, ENSAE, IPP, Palaiseau, France
- 2003–2006 **Lecturer in Statistics**, Bristol University, UK
- 2002–2003 **Statistical methodologist**, UMS-INSEE, Paris, France
- 1999–2002 **Junior Lecturer in Statistics**, ENSAE, Malakoff, France

Education

- 2010 **Habilitation à Diriger des recherches**, Université Paris Dauphine
(French habilitation to supervise Ph.D. students, Viva 1st Dec. 2010)
- 1999–2003 **Ph.D. thesis in Statistics**, Université Paris VI, supervisor: Pr C.P. Robert
Sequential Monte Carlo methods and their applications to Bayesian inference.
- 1997–1999 **Msc. in Economics and Statistics**, ENSAE
- 1994–1997 **Msc. in Engineering**, Ecole Polytechnique

Research Interests

I am interested in Bayesian computation in a general sense, that is, the development of numerical methods to perform Bayesian learning and inference. I am particularly interested in:

- Monte Carlo methods, especially Sequential Monte Carlo (particle filtering): theory (convergence properties, etc), methodology, and applications; Importance sampling, Markov chain Monte Carlo (MCMC), and quasi-Monte Carlo.
- Deterministic approximations: Laplace, Variational Bayes and Expectation Propagation.

I am also interested in Bayesian modelling and practical applications: I am collaborating or have collaborated with scientists from different areas of Science, such as cosmology, neuro-sciences, genetics, and public health.

Awards and distinctions

- **Keynote speaker**: MCM 2025 (Chicago), ISBA 2018 (Edinburgh), MCQMC 2016 (Stanford)
- 2021 de Groot award for best published book in Statistics and decision theory, International Society for Bayesian Analysis.
- Fellow of the IMS (Institute of Mathematical Statistics), 2020
- 2002 Leonard J. Savage Award for an outstanding doctoral dissertation in Bayesian econometrics and statistics ('Theory and Methods' section), International Society for Bayesian Analysis.

Grants

- ANR grant BLISS, 2024-2027, 363 KE; ""Beyond Linearization: Automated Inference for Nonlinear State-Space Models" (joint with PI Andras Fulop, ESSEC and co-PI Anna Simoni, CREST)
- ANR grant B3DCMB, 2017-2021, 531 KE; "Big Bang from Big CMB Data" (joint with PI Radek Stompor, CNRS, and co-PI Laura Grigori, INRIA)
- ANR grant "SP Bayes", 2008-2010 (Statistique Bayésienne semi-paramétrique) joint with PI Judith Rousseau
- EPSRC first grant EP/C015886/1, 2005-2008, 74 K£; 'Novel Applications of Sequential Monte Carlo'.

Teaching

- **ENSAE**: "Introduction to Stochastic processes" (2nd year, 20 hrs), "Statistique 1" (2nd year, 20 hrs, 2006-2018), "Monte Carlo and simulation" (2nd year, 16 hrs), "State-space models and sequential Monte Carlo (3rd year, 18 hrs)
- **Master M2 MVA**: "Probabilistic Graphical models" (M2, half of the 24h course)
- **Université Paris Dauphine**: "Hidden-Markov models and Particle methods" (M2, 18 hours, 2007 to 2015); "Time Series" (M1, 20 hours, 2008-09)
- **HEC Paris**: "Statistique" (L3, 50 hrs, 2011-2017)
- **HEC Lausanne**: "Introduction to Time series" (Msc, 28 hours, 2009-2014)
- **Bristol University**: Bayesian Analysis (18 hours, lvl 3, 2003-06), Introduction to Statistics (18 hours, lvl 2, 2003-06)
- **Ecole Polytechnique**: "chargé de cours incomplet" (60 hours, 2007-08)
- **Università dell'Insubria**: Bayesian Analysis (18 hours, 2005)

Academic Responsibilities

- **Editor**: J. Roy. Stat. Soc. B (2026-2028).
- **Associate Editor**: Annals of Statistics (2019-2025); Biometrika (2018-2025); J. Roy. Stat. Soc. B (2012-2020); Stats & Comp. (2013-2017); Stat. Methods and Appl. (2012-2015); member (2013-14) and secretary (2015-16) of the research section of the RSS.
- **Membership**: American Statistical Association (ASA), Inst. of Mathematical Statistics (IMS), Royal Statistical Society (RSS), International Society for Bayesian Analysis (ISBA).
- **Post-doc supervision**: Andrea Pandolfi (2025-27), Francesca Crucinio (2023).
- **Ph.D. supervision**: Aleksei Arzhantsev (2026-29), Yuedan Huo (2025-28), Yvann Le Fay (2024-27), Sylvain Procope-Mamert (2023-26), Otmane Sakhi (2020-23, Criteo), Younès Youssfi (2020-23), Hai-Dang Dau (2019-22), Gabriel Ducrocq (2019-22), Lionel Riou-Durand (2016-19, post-doc at Warwick U), Alexander Buchholz (2015-18, Amazon), Charles Findling (2015-18, co-supervision at ENS), Vincent Cottet (2014-17, French administration), Mathieu Gerber (2012-15, Bristol University), James Ridgway (2012-15, Capital Fund Management), Pierre Jacob (2009-12, co-supervision, assistant prof at Havard U), Christian Schäfer (2009-2012, Dept), Giusi Moffa (Bristol, 2005-06), Elisa Varini (Bristol, 2004-05, co-supervision, CNR)

- **Ph.D. examiner:** Eleonore Blanchard (2026), David Xu (UBC, 2025), Andrea Pandolfi (Bocconi, 2025), Luc de Montella (INRIA Bordeaux, 2025), Jacopo Iollo (INRIA Grenoble, 2025), Antoine Picard (INRIA Lille, 2025), Charly Andral (Dauphine, 2024), Roberta Flenghi (Les Ponts, 2023), Francesca Crucinio (Warwick, 2021), Tobias Schwedes (Imperial, 2019), Jordan Franks (Jyväskylä, 2019), Daniel W. Kennedy (QUT, 2019), Changye Wu (Dauphine, 2018), Jeremy Heng (Oxford, 2016), Van Bien Bui (Nice, 2016), Damien Juery (Montpellier, 2014), Yohan Petetin (Télécom Sud-Paris, 2014), Paul Bui Quang (Rennes, 2013), Sinan Yildirim (Cambridge, 2012), Salima El Kolei (Nice, 2012), Mohamed Sedki (Montpellier, 2012), Anthony Lee (Oxford, 2011), Meïli Baragatti (Marseille, 2011), Nicole White (QUT, 2011), Hugo Hammer (Trondheim, 2008), Zhen Liu (Lancaster, 2008), Adam Johansen (Cambridge, 2006), David Hastie (Bristol, 2004).
- **Conference organisation:** Master class in Bayesian Statistics (CIRM, Oct 2018), SMC 2015 (Paris, 26-28 Aug), NeuroStats 2014 (Warwick, 3-5 Sept)
- **Visits:** Postdam (Aug 2022), Yale (Feb 2019), Oxford (Apr 2017), Pompeu Fabra, Barcelone (May 2013), INRIA Bordeaux (Jul 2011), Cambridge (Oct 2010, Jan 2008, Apr 2007), Lisboa (Nov 2010), Insubria (Nov 2005), Trondheim (March 2013, Sept 2005), Bank of Canada (Sept 2004).

Book

- Chopin, N. and Papaspiliopoulos, O. (2020). An introduction to Sequential Monte Carlo, Springer, doi: 10.1007/978-3-030-47845-2.

Publications

- [1] C. ANDRIEU, N. CHOPIN, E. FINCATO, and M. GERBER. Convergence of a class of gradient-free optimisation schemes when the objective function is noisy, irregular, or both. **arxiv 2512.03225** (Dec. 2025).
- [2] Y. L. FAY, N. CHOPIN, and S. BARTHELMÉ. Least squares variational inference. **Advances in Neural Information Processing Systems** (2025).
- [3] N. CHOPIN, H. WANG, and M. GERBER. Adaptive stratified Monte Carlo using decision trees. **Stat. Comput.** 35.193 (2025).
- [4] A. CORENFLOS, M. SUTTON, and N. CHOPIN. Debiasing piecewise deterministic Markov process samplers using couplings. **Scand. J. Stat.** 52.4 (2025), 1932–1974.
- [5] M. XI, Z. SHEN, M. RIABIZ, N. CHOPIN, and C. J. OATES. Extrapolation of tempered posteriors. **arxiv 2509.12173** (Sept. 2025).
- [6] N. CHOPIN, F. R. CRUCINIO, and S. S. SINGH. Towards a turnkey approach for unbiased Monte Carlo estimation of smooth functions of expectations. **Biometrika** 112.3 (2025).
- [7] T. VOLDOIRE, N. CHOPIN, G. RATEAU, and R. J. RYDER. Saddlepoint Monte Carlo and its application to exact ecological inference. **arxiv 2410.18243** (Oct. 2024).
- [8] C. ANDRIEU, N. CHOPIN, E. FINCATO, and M. GERBER. Gradient-free optimization via integration. **arxiv 2408.00888** (Aug. 2024).
- [9] O. SAKHI, I. AOUALI, P. ALQUIER, and N. CHOPIN. Logarithmic Smoothing for Pessimistic Off-Policy Evaluation, Selection and Learning. **Advances in Neural Information Processing Systems** 37 (2024), 80706–80755.
- [10] N. CHOPIN, F. R. CRUCINIO, and A. KORBA. A connection between tempering and entropic mirror descent. **Forty-first International Conference on Machine Learning**. 2024.
- [11] N. CHOPIN and M. GERBER. Higher-order Monte Carlo through cubic stratification. **SIAM J. Numer. Anal.** 62.1 (2024), 229–247.
- [12] Y. YOUSSEFI and N. CHOPIN. Scalable Bayesian bi-level variable selection in generalized linear models. **Foundations of Data Science** 7.4 (2024), 1020–1033.

- [13] R. JIN, S. S. SINGH, and N. CHOPIN. De-biasing particle filtering for a continuous time hidden Markov model with a Cox process observation model. **Statist. Sinica** 34 (2024), 1215–1239.
- [14] O. SAKHI, D. ROHDE, and N. CHOPIN. Fast Slate Policy Optimization: Going Beyond Plackett-Luce. **Transactions on Machine Learning Research** (2023).
- [15] O. SAKHI, P. ALQUIER, and N. CHOPIN. PAC-Bayesian Offline Contextual Bandits With Guarantees. **Proceedings of the 40th International Conference on Machine Learning**. Ed. by A. KRAUSE et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 23–29 Jul 2023, 29777–29799.
- [16] H.-D. DAU and N. CHOPIN. On backward smoothing algorithms. **Ann. Statist.** 51.5 (2023), 2145–2169.
- [17] N. CHOPIN, A. FULOP, J. HENG, and A. H. THIERY. Computational Doob h-transforms for Online Filtering of Discretely Observed Diffusions. **Proceedings of the 40th International Conference on Machine Learning**. Ed. by A. KRAUSE et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 23–29 Jul 2023, 5904–5923.
- [18] N. CHOPIN, S. S. SINGH, T. SOTO, and M. VIHOLA. On resampling schemes for particle filters with weakly informative observations. **Ann. Statist.** 50.6 (2022), 3197–3222.
- [19] A. CORENFLOS, N. CHOPIN, and S. SÄRKKÄ. De-Sequentialized Monte Carlo: a parallel-in-time particle smoother. **Journal of Machine Learning Research** 23.283 (2022), 1–39.
- [20] G. DUCROCQ, N. CHOPIN, J. ERRARD, and R. STOMPOR. Improved Gibbs samplers for cosmic microwave background power spectrum estimation. **Phys. Rev. D** 105 (10 May 2022), 103501.
- [21] H.-D. DAU and N. CHOPIN. Waste-free sequential Monte Carlo. **J. R. Stat. Soc. Ser. B. Stat. Methodol.** 84.1 (2022), 114–148.
- [22] N. CHOPIN and G. DUCROCQ. Fast compression of MCMC output. **Entropy** 23.8 (2021), Paper No. 1017, 16.
- [23] A. BUCHHOLZ, N. CHOPIN, and P. E. JACOB. Adaptive tuning of Hamiltonian Monte Carlo within sequential Monte Carlo. **Bayesian Anal.** 16.3 (2021), 745–771.
- [24] D. ALVARES, C. ARMERO, A. FORTE, and N. CHOPIN. Sequential Monte Carlo methods in Bayesian joint models for longitudinal and time-to-event data. **Stat. Model.** 21.1-2 (2021), 161–181.
- [25] C. FINDLING, N. CHOPIN, and E. KOECHLIN. Imprecise neural computations as a source of adaptive behaviour in volatile environments. **Nature Human Behaviour** (2020), 1–14.
- [26] M. GERBER, N. CHOPIN, and N. WHITELEY. Negative association, ordering and convergence of resampling methods. **Ann. Statist.** 47.4 (2019), 2236–2260.
- [27] A. BUCHHOLZ and N. CHOPIN. Improving Approximate Bayesian Computation via Quasi-Monte Carlo. **J. Comput. Graph. Statist.** 28.1 (2019), 205–219.
- [28] S. BARTHELMÉ, N. CHOPIN, and V. COTTET. Divide and conquer in ABC: expectation-propagation algorithms for likelihood-free inference. **Handbook of approximate Bayesian computation**. Chapman & Hall/CRC Handb. Mod. Stat. Methods. CRC Press, Boca Raton, FL, 2019, 415–434.
- [29] C. ANDRIEU, A. DOUCET, S. YILDIRIM, and N. CHOPIN. On the utility of Metropolis-Hastings with asymmetric acceptance ratio. **arXiv e-prints**, arXiv:1803.09527 (Mar. 2018).
- [30] L. RIOU-DURAND and N. CHOPIN. Noise contrastive estimation: asymptotic properties, formal comparison with MC-MLE. **Electron. J. Stat.** 12.2 (2018), 3473–3518.
- [31] N. CHOPIN and M. GERBER. Sequential quasi-Monte Carlo: introduction for non-experts, dimension reduction, application to partly observed diffusion processes. **Monte Carlo and quasi-Monte Carlo methods**. Vol. 241. Springer Proc. Math. Stat. Springer, Cham, 2018, 99–121.
- [32] D. ALVARES, C. ARMERO, A. FORTE, and N. CHOPIN. Sequential Monte Carlo methods in random intercept models for longitudinal data. **Bayesian statistics in action**. Vol. 194. Springer Proc. Math. Stat. Springer, Cham, 2017, 3–9.

- [33] S. VASISHTH, N. CHOPIN, R. RYDER, and B. NICENBOIM. Modelling dependency completion in sentence comprehension as a Bayesian hierarchical mixture process: A case study involving Chinese relative clauses. **Proceedings of the 39th annual meeting of the cognitive science society**. July 2017.
- [34] S. VASISHTH, B. NICENBOIM, N. CHOPIN, and R. RYDER. Bayesian Hierarchical Finite Mixture Models of Reading Times: A Case Study. **PsyArXiv** (July 2017).
- [35] C. J. OATES, M. GIROLAMI, and N. CHOPIN. Control functionals for Monte Carlo integration. **J. R. Stat. Soc. Ser. B. Stat. Methodol.** 79.3 (2017), 695–718.
- [36] M. GERBER and N. CHOPIN. Convergence of sequential quasi-Monte Carlo smoothing algorithms. **Bernoulli** 23.4B (2017), 2951–2987.
- [37] N. CHOPIN and J. RIDGWAY. Leave Pima Indians alone: binary regression as a benchmark for Bayesian computation. **Statist. Sci.** 32.1 (2017), 64–87.
- [38] C. SCHRETTTER, Z. HE, M. GERBER, N. CHOPIN, and H. NIEDERREITER. Van der Corput and golden ratio sequences along the Hilbert space-filling curve. **Monte Carlo and quasi-Monte Carlo methods**. Vol. 163. Springer Proc. Math. Stat. Springer, [Cham], 2016, 531–544.
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- [40] N. CHOPIN and M. GERBER. Application of sequential Quasi-Monte Carlo to autonomous positioning. **Signal Processing Conference (EUSIPCO), 2015 23rd European**. Aug. 2015, 489–493.
- [41] N. CHOPIN, J. RIDGWAY, M. GERBER, and O. PAPASPILIOPOULOS. Towards automatic calibration of the number of state particles within the SMC² algorithm. **ArXiv preprint 1506.00570** (June 2015).
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- [43] M. GERBER and N. CHOPIN. Sequential quasi Monte Carlo. **J. R. Stat. Soc. Ser. B. Stat. Methodol.** 77.3 (2015), 509–579.
- [44] S. BARTHELMÉ and N. CHOPIN. The Poisson transform for unnormalised statistical models. **Stat. Comput.** 25.4 (2015), 767–780.
- [45] N. CHOPIN and S. S. SINGH. On particle Gibbs sampling. **Bernoulli** 21.3 (2015), 1855–1883.
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- [51] C. ANDRIEU, N. CHOPIN, A. DOUCET, and S. RUBENTHALER. Perfect simulation for the Feynman-Kac law on the path space. **ArXiv preprint 1210.0376** (Mar. 2013).
- [52] N. CHOPIN, P. E. JACOB, and O. PAPASPILIOPOULOS. SMC²: an efficient algorithm for sequential analysis of state space models. **J. R. Stat. Soc. Ser. B. Stat. Methodol.** 75.3 (2013), 397–426.
- [53] N. CHOPIN, J. ROUSSEAU, and B. LISEO. Computational aspects of Bayesian spectral density estimation. **J. Comput. Graph. Statist.** 22.3 (2013), 533–557.

- [54] S. S. SINGH, N. CHOPIN, and N. WHITELEY. Bayesian learning of noisy Markov decision processes. **ACM Trans. Model. Comput. Simul.** 23.1 (2013), Art. 4, 25.
- [55] C. SCHÄFER and N. CHOPIN. Sequential Monte Carlo on large binary sampling spaces. **Stat. Comput.** 23.2 (2013), 163–184.
- [56] N. CHOPIN, A. GELMAN, K. L. Mengersen, and C. P. ROBERT. In praise of the referee. **ArXiv preprint 1205.4304** (May 2012).
- [57] C. ANDRIEU et al. Some discussions of D. Fearnhead and D. Prangle's Read Paper "Constructing summary statistics for approximate Bayesian computation: semi-automatic approximate Bayesian computation". **ArXiv preprint 1201.1314** (Jan. 2012).
- [58] N. CHOPIN and C. ROBERT. Discussion of "Catching up faster by switching sooner: a predictive approach to adaptive estimation with an application to the AIC–BIC dilemma" by Erven, Tim van and Grünwald, Peter and de Rooij, Steven. **Journal of the Royal Statistical Society (series B)** 74.3 (2012), 361–417.
- [59] J. ROUSSEAU, N. CHOPIN, and B. LISEO. Bayesian nonparametric estimation of the spectral density of a long or intermediate memory Gaussian process. **Ann. Statist.** 40.2 (2012), 964–995.
- [60] N. CHOPIN, T. LELIÈVRE, and G. STOLTZ. Free energy methods for Bayesian inference: efficient exploration of univariate Gaussian mixture posteriors. **Stat. Comput.** 22.4 (22 2012), 897–916.
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- [63] N. CHOPIN and O. PAPASPILIOPOULOS. Comments on "Bayesian variable selection for random intercept modeling of Gaussian and non-Gaussian Data" by Frühwirth-Schnatter, S. and Wagner, H. **Bayesian Statistics 9**. Ed. by J. M. BERNARDO et al. Oxford University Press, 2011, 165–200.
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- [65] N. CHOPIN, P. DEL MORAL, and S. RUBENTHALER. Stability of Feynman-Kac formulae with path-dependent potentials. **Stochastic Process. Appl.** 121.1 (2011), 38–60.
- [66] N. CHOPIN et al. On Particle Learning; comments on "Particle learning for sequential Bayesian computation" by Lopes, Carvalho, Johannes, and Polson. **Bayesian Statistics 9**. Ed. by J. M. BERNARDO et al. Oxford University Press, 2011, 317–360.
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- [69] C. P. ROBERT, N. CHOPIN, and J. ROUSSEAU. Rejoinder: Harold Jeffreys's Theory of Probability Revisited. **ArXiv preprint 0909.1008** (Jan. 2010).
- [70] N. CHOPIN and C. P. ROBERT. Properties of nested sampling. **Biometrika** 97.3 (2010), 741–755.
- [71] P. JACOB, N. CHOPIN, C. P. ROBERT, and H. RUE. Comments on "Particle Markov chain Monte Carlo" by C. Andrieu, A. Doucet, and R. Holten. **ArXiv preprint 0911.0985** (Nov. 2009).
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- [79] N. CHOPIN. Comment on 'Sequential Monte Carlo for Bayesian computation' by Del Moral et al. **Bayesian Statistics 8**. Ed. by O. U. P. BERNARDO J. M. ET AL. (EDS). 2007, 115–148.
- [80] H. RUE, S. MARTINO, and N. CHOPIN. Discussion on 'Modern Statistics for Spatial Point Processes' by Møller and Waagepetersen. **Scandinavian Journal of Statistics** 34.4 (2007), 685–711.
- [81] N. CHOPIN. Dynamic detection of change points in long time series. **Ann. Inst. Statist. Math.** 59.2 (2007), 349–366.
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